Student Performance Analysis

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About

This project involves the analysis of student performance data, which helps in the identification of the major factors affecting the performance of the students. In this regard, the analysis is carried out using SQL for data exploration and Power BI for data visualization, which helps in understanding the impact of several factors, such as the number of study hours, the extent of sleep, class attendance, and levels of motivation, on the scores obtained by the students. The analysis is also carried out with the objective of identifying the risk among the students.

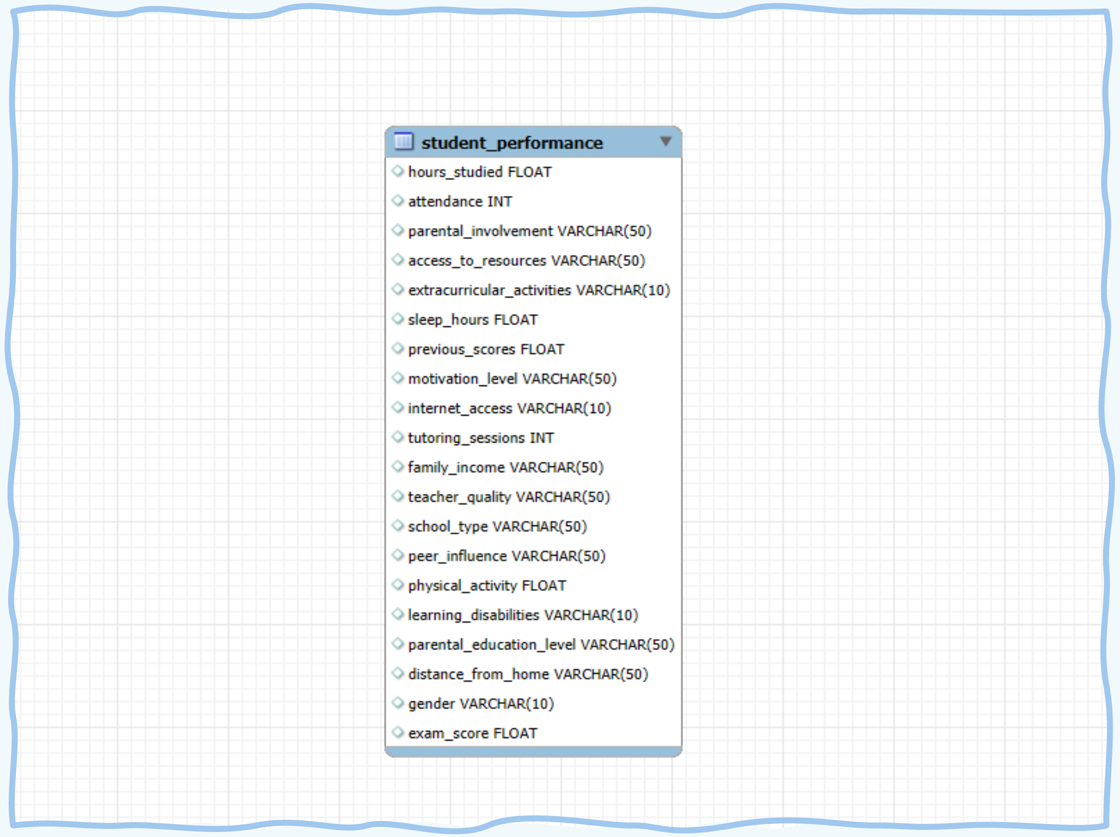
Our goals

To identify the key factors (study hours, sleep, attendance, motivation) that influence student performance

To analyze patterns and characteristics of high-performing students

To identify at-risk students and provide actionable insights to improve academic outcomes

Analysis using SQL



ERR Diagram of the Data

Overall Performance

Q1. What is the **average, max, and min exam score**?

Query:

```
SELECT AVG(exam_score) AS Avg,
MIN(exam_score) AS Min,
MAX(exam_score) AS Max
FROM student_performance;
```

Answer and Analysis:

Avg	Min	Max
67.23565914938702	55	101

Here, the average score of the students is 67.235
The minimum and the maximum scores of the students are 55 and 101 respectively, suggesting a relatively wide distribution in student performance. This variation highlights the presence of both low-performing and high-performing students, making it important to further analyze the factors driving these differences.

Attendance Impact

Q2. What is the **average score** based on **attendance ranges**?

Query:

```
SELECT CASE
        WHEN attendance <= 60 THEN 'Below 60'
        WHEN attendance BETWEEN 60 AND 80 THEN '60-80'
        ELSE 'Above 80'
    END AS attendance_group,
    AVG(exam_score) AS avg_score FROM student_performance
GROUP BY attendance_group;
```

Answer and Analysis:

attendance_group	avg_score
Above 80	69.24434673366834
60-80	65.44424460431655
Below 60	62.41379310344828

In this table, The students have been grouped according to their levels of attendance. From the observation, the students who have more than 80% attendance have the highest scores compared to the others. There is also a declining pattern of scores as the attendance levels decrease.

This implies there is a positive correlation between the students' attendance and their scores. This means that the more the students attend, the higher the scores they attain. However, the students' attendance should not be the only aspect considered.

Study Hours Effect

Q3. Do more **study hours** lead to higher scores?

Query:

```
SELECT CASE
        WHEN hours_studied <= 10 THEN 'Below 10'
        WHEN hours_studied BETWEEN 10 AND 20 THEN '10-20'
        ELSE 'Above 20'
    END AS Hours_studied_data,
    AVG(exam_score) AS avg_score FROM student_performance
GROUP BY hours_studied_data;
```

Answer and Analysis:

Hours_studied_data	avg_score
Above 20	68.7231472412667 3
10-20	66.1777288455489 1
Below 10	63.967123287671235

Here, The students were grouped into three categories based on the number of study hours: less than 10, 10-20, and more than 20. The analysis indicates that there is an upward trend in the average scores as the number of study hours increases.

The average score of the students studying more than 20 hours was the highest, at 68.72, followed by those studying between 10-20 hours, with an average score of 66.17. The average score of the students studying less than 10 hours was the lowest, at 63.96.

Top vs Bottom Students

Q4. Compare top scorers and the bottom scorers. What factors differ?

Query:

```sql
SELECT
    CASE
        WHEN exam_score >= 75 THEN 'Top Students'
        WHEN exam_score <= 60 THEN 'Bottom Students'
    END AS student_group,
    AVG(hours_studied) AS avg_hours,
    AVG(attendance) AS avg_attendance,
    AVG(sleep_hours) AS avg_sleep FROM student_performance
WHERE exam_score >= 75 OR exam_score <= 60
GROUP BY student_group;
```

Answer and Analysis:

student_group	avg_hours	avg_attendance	avg_sleep
Bottom Students	11.6	64.8069	7.068965517241379
Top Students	26.266129032258064	88.4597	6.879032258064516

Here, lets consider the students who has scored 75 and more as ‘Top Students’ and the ones who has scored less than 60 as ‘Bottom Students’ as the average is 67. Here, we have taken factors like Average study hours, average attendance and Average sleep of students to check the differences.

In this table, the bottom students can be seen having:
- Less average study hour compared to top students
- Less average attendance
- Better average sleep hours

We can come into conclusion saying that the bottom students do not prioritize study hours or even attendance when compared to top students.

Parental Influence

Q5. Does **parental education level** affect performance?

Query:

```
SELECT parental_education_level,
COUNT(*) AS count_students FROM (
    SELECT parental_education_level
    FROM student_performance
    ORDER BY exam_score DESC
    LIMIT 50
) AS top_students
GROUP BY parental_education_level
ORDER BY count_students DESC;
```

Answer and Analysis:

parental_education_level	count_students
High School	24
College	16
Postgraduate	10

Here, The education level of the parents of the top 50 students, based on their exam scores, was studied. The data indicates that the majority of the high-performing students come from parents with a high school level education, compared to those with a college or postgraduate background.

This suggests that the education level of the parents does not have a major impact on the performance of the student, which implies that other factors, like the efforts of the individual, might be more crucial.

Sleep vs Performance

Q6. Is there an **optimal sleep range** for best scores?

Query:

```
SELECT CASE WHEN sleep_hours < 5 THEN 'Less than 5'
WHEN sleep_hours BETWEEN 5 AND 8 THEN '5-8'
ELSE 'More than 8' END AS SLEEP_RANGE,
AVG(exam_score) AS avg_score FROM student_performance
GROUP BY SLEEP_RANGE
ORDER BY avg_score DESC;
```

Answer and Analysis:

SLEEP_RANGE	avg_score
Less than 5	67.62783171521036
5-8	67.2310497025523
More than 8	67.14627414903404

In this table, The students were then divided into three categories of sleep duration: less than 5 hours, 5-8 hours, and more than 8 hours. This analysis reveals that the average scores of the students in the three categories are more or less the same, with minor variations.

This reveals that the duration of sleep for the students has no impact on their performance in the exams. However, it might have an impact in association with other factors such as the way of studying and routine.

Key Success Factors

Q7. Among high scorers (say score > 80), what patterns do you see?

Query:

```
SELECT AVG(hours_studied) AS avg_hours_studied,
AVG(attendance) AS avg_attendance,
AVG(sleep_hours) AS avg_sleep
FROM student_performance
WHERE exam_score > 80;
```

Answer and Analysis:

avg_hours_studied	avg_attendance	avg_sleep
19.651162790697676	80.0698	6.930232558139535

For high-performing students, i.e., students who scored more than 80, it was observed that they studied for an average of around 19.6 hours, indicating the importance of studying regularly for achieving high scores. The attendance for high-performing students was also around 80%.

Furthermore, the high-performing students slept for an average of around 7 hours, indicating the importance of maintaining a proper routine instead of extreme study habits. It was observed that high scores are the result of regular study, regular attendance, and proper sleep.

Risk Group Identification

Q8. Identify students who has low attendance, low study hours, low scores (At-risk students)

Query:

```
#A: Look into Average, Min and Max:

• SELECT MIN(attendance), MAX(attendance), AVG(attendance), MIN(hours_studied), MAX(hours_studied), AVG(hours_studied), MIN(exam_score), MAX(exam_score), AVG(exam_score)
  FROM student_performance;


#B: Take AVG as benchmark and classify:

• SELECT COUNT(*) AS at_risk_students FROM student_performance
  WHERE attendance < 79
  AND hours_studied < 19
  AND exam_score < 67;


#C: Take total number and compare:

• SELECT COUNT(*) AS total_students FROM student_performance;
```

Risk Group Identification

Answer and Analysis:

MIN(attendance)	MAX(attendance)	AVG(attendance)	MIN(hours_studied)	MAX(hours_studied)	AVG(hours_studied)	MIN(exam_score)	MAX(exam_score)	AVG(exam_score)
60	100	79.9774	1	44	19.975329196306948	55	101	67.23565914938702

Here, we have taken the average, minimum and maximum of attendance, study hours and exam scores of the students to identify the overall performance and to set a benchmark.

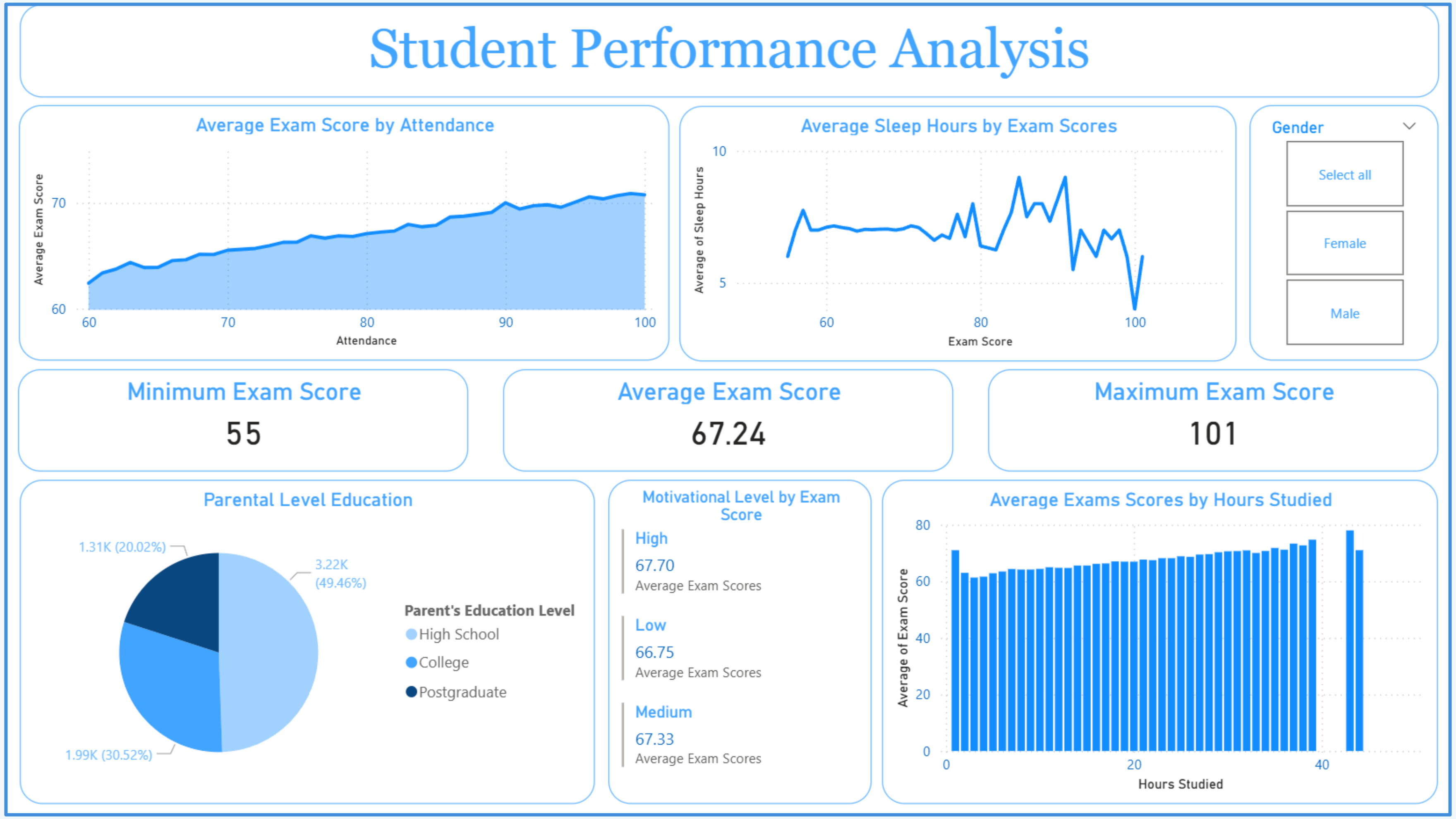
at_risk_students
1118

Taking the average as the benchmark, we got the students who have scored less than average, have less study hours and also the ones who have less attendance. So now, we can classify those group of students as At-risk students.

total_students
6607

To compare the number or the ratio of At-risk students, we can take the total number of students and then compare them together. This gives us that 16.9 percent of the students are considered as being at-risk, i.e., being below average.

Power BI Analysis (Dashboard)



Key Takeaways

Attendance Strongly Drives Performance	Students with above 80% attendance achieve the highest average exam scores (69.24), showing a clear positive correlation between regular class presence and better academic outcomes
Study Hours Deliver Measurable Gains	More study hours consistently lead to higher scores, with students studying over 20 hours posting the top average of 68.72 compared to 63.96 for those studying under 10 hours.
Sleep Duration Has Negligible Impact	Average exam scores remain nearly identical across sleep ranges (less than 5, 5-8, or more than 8 hours), indicating that sleep duration alone does not significantly affect performance.
Top Scorers Follow Balanced Habits	High-performing students (score >80) average 19.6 study hours, 80% attendance, and 7 hours of sleep, proving that consistent routines—not extreme efforts—drive success.
Parental Education Level Is Not a Major Factor	Among the top 50 students, the majority come from high-school-educated parents, showing that individual effort matters far more than parents' education background.
16.9% of Students Are At-Risk	Using average benchmarks, 1,118 students (16.9% of the total) show low attendance (<79%), low study hours (<19), and below-average scores (<67), highlighting a clear group needing targeted support.

Reccomendation

1) Enhance School Attendance to Boost Academic Achievement
The best method to increase student success is through increasing the amount of time spent outside of the home. The most significant increase in test results(average increase of 69.24%) occurred with students who attended school consistently for 80% of the time.

2) Promote Consistent Studying
Students should encourage themselves to set time aside each week for studying (15-20+ hours); this will reduce the last-minute cramming that accompanies the student studying more than once after multiple hours of studying. Greater amounts of studying lead to higher average grades.

3) Identify at-Risk Students for Early Intervention
Use easy-to-interpret data points to identify approximately 16.9% of students who are at risk based on <79% attendance rate or <19 hours of study per week or scores <67; these students should be provided additional mentor support, money for classes and regular evaluation of their own efforts.

4) Focus on Creating a Routine
Students should try and get 7-8 amount of sleep per night and regularly attend classes instead of attempting to complete multiple hours of study in a single weekend. The highest achievers study on a consistent schedule as opposed to being drained from studying for extended amounts of time prior to testing.

5) Remove Parent’s Education Level from School Support
Schools/counselors should provide the same level of resources to support all students regardless of their parent's education, as the effort of the student is worth much more than the school parent's educational attainment.

Conclusion

This analysis unequivocally demonstrates that while sleep duration and parental education have little to no influence on student performance, controllable habits—regular attendance and consistent study hours—have a major influence. Schools can greatly enhance academic results and assist every student in realizing their full potential by implementing these data-driven recommendations and providing support to the 16.9% of at-risk students.

Thank You